

$$A_i = \begin{bmatrix} \cos \theta_i & -\sin \theta_i \cos \alpha_i & \sin \theta_i \sin \alpha_i & a_i \cos \theta_i \\ \sin \theta_i & \cos \theta_i \cos \alpha_i & -\cos \theta_i \sin \alpha_i & a_i \sin \theta_i \\ 0 & \sin \alpha_i & \cos \alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

i	θ_i	d_i [mm]	a_i [mm]	α_i
1	q_1	89.45	0	$\frac{\pi}{2}$
2	q_2	0	105.95	0
3	q_3	0	100	0
4	q_4	0	0	$\frac{\pi}{2}$

$$c_i = \cos q_i, \quad s_i = \sin q_i$$

$$A_1 = \begin{bmatrix} c_1 & -s_1 \cos\left(\frac{\pi}{2}\right) & s_1 \sin\left(\frac{\pi}{2}\right) & 0 \\ s_1 & c_1 \cos\left(\frac{\pi}{2}\right) & -c_1 \sin\left(\frac{\pi}{2}\right) & 0 \\ 0 & \sin\left(\frac{\pi}{2}\right) & \cos\left(\frac{\pi}{2}\right) & 89.45 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_1 = \begin{bmatrix} c_1 & 0 & s_1 & 0 \\ s_1 & 0 & -c_1 & 0 \\ 0 & 1 & 0 & 89.45 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} c_2 & -s_2 \cos(0) & s_2 \sin(0) & 105.95c_2 \\ s_2 & c_2 \cos(0) & -c_2 \sin(0) & 105.95s_2 \\ 0 & \sin(0) & \cos(0) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} c_2 & -s_2 & 0 & 105.95c_2 \\ s_2 & c_2 & 0 & 105.95s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} c_3 & -s_3 \cos(0) & s_3 \sin(0) & 100c_3 \\ s_3 & c_3 \cos(0) & -c_3 \sin(0) & 100s_3 \\ 0 & \sin(0) & \cos(0) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} c_3 & -s_3 & 0 & 100c_3 \\ s_3 & c_3 & 0 & 100s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_4 = \begin{bmatrix} c_4 & -s_4 \cos\left(\frac{\pi}{2}\right) & s_4 \sin\left(\frac{\pi}{2}\right) & 0 \\ s_4 & c_4 \cos\left(\frac{\pi}{2}\right) & -c_4 \sin\left(\frac{\pi}{2}\right) & 0 \\ 0 & \sin\left(\frac{\pi}{2}\right) & \cos\left(\frac{\pi}{2}\right) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_4 = \begin{bmatrix} c_4 & 0 & s_4 & 0 \\ s_4 & 0 & -c_4 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0T_4 = A_1 A_2 A_3 A_4$$

$$c_{23} = \cos(q_2 + q_3), \quad s_{23} = \sin(q_2 + q_3)$$

$$c_{234} = \cos(q_2 + q_3 + q_4), \quad s_{234} = \sin(q_2 + q_3 + q_4)$$

$${}^0T_4 = \begin{bmatrix} c_1 c_{234} & s_1 & c_1 s_{234} & c_1 (105.95c_2 + 100c_{23}) \\ s_1 c_{234} & -c_1 & s_1 s_{234} & s_1 (105.95c_2 + 100c_{23}) \\ s_{234} & 0 & -c_{234} & 89.45 + 105.95s_2 + 100s_{23} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\rho = 105.95 \cos q_2 + 100 \cos(q_2 + q_3)$$

$$x = \rho \cos q_1$$

$$y = \rho \sin q_1$$

$$z = 89.45 + 105.95 \sin q_2 + 100 \sin(q_2 + q_3)$$

$$R_4^0 = \begin{bmatrix} c_1 c_{234} & s_1 & c_1 s_{234} \\ s_1 c_{234} & -c_1 & s_1 s_{234} \\ s_{234} & 0 & -c_{234} \end{bmatrix}$$

$$\text{roll} = \text{atan2}(r_{32}, r_{33}) = \text{atan2}(0, -c_{234})$$

$$\text{pitch} = \text{atan2}\left(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2}\right)$$

$$\text{pitch} = \text{atan2} \left(-s_{234}, \sqrt{c_1^2 c_{234}^2 + s_1^2 c_{234}^2} \right)$$

$$\text{yaw} = \text{atan2}(r_{21}, r_{11}) = \text{atan2}(s_1 c_{234}, c_1 c_{234})$$

$$\cos(q_2 + q_3 + q_4) > 0$$

$$\begin{bmatrix} \text{roll} \\ \text{pitch} \\ \text{yaw} \end{bmatrix} = \begin{bmatrix} 180^\circ \\ -(q_2 + q_3 + q_4) \\ q_1 \end{bmatrix}$$

$$\mathbf{q}^{(1)} = \begin{bmatrix} 0^\circ & 0^\circ & 0^\circ & 0^\circ \end{bmatrix}^T$$

$$\rho_1 = 105.95 \cos(0^\circ) + 100 \cos(0^\circ + 0^\circ) = 205.950000 \text{ mm}$$

$$x_1 = 205.95 \cos(0^\circ) = 205.950000 \text{ mm} = 0.205950 \text{ m}$$

$$y_1 = 205.95 \sin(0^\circ) = 0 \text{ mm} = 0 \text{ m}$$

$$z_1 = 89.45 + 105.95 \sin(0^\circ) + 100 \sin(0^\circ + 0^\circ) = 89.450000 \text{ mm} = 0.089450 \text{ m}$$

$$\begin{bmatrix} \text{roll}_1 \\ \text{pitch}_1 \\ \text{yaw}_1 \end{bmatrix} = \begin{bmatrix} 180^\circ \\ -(0^\circ + 0^\circ + 0^\circ) \\ 0^\circ \end{bmatrix} = \begin{bmatrix} 180^\circ \\ 0^\circ \\ 0^\circ \end{bmatrix}$$

$$\mathbf{q}^{(2)} = \begin{bmatrix} 25^\circ & 25^\circ & 20^\circ & -20^\circ \end{bmatrix}^T$$

$$\rho_2 = 105.95 \cos(25^\circ) + 100 \cos(25^\circ + 20^\circ) = 166.733988 \text{ mm}$$

$$x_2 = 166.733988 \cos(25^\circ) = 151.112312 \text{ mm} = 0.151112 \text{ m}$$

$$y_2 = 166.733988 \sin(25^\circ) = 70.464828 \text{ mm} = 0.070465 \text{ m}$$

$$z_2 = 89.45 + 105.95 \sin(25^\circ) + 100 \sin(25^\circ + 20^\circ) = 204.937083 \text{ mm} = 0.204937 \text{ m}$$

$$\begin{bmatrix} \text{roll}_2 \\ \text{pitch}_2 \\ \text{yaw}_2 \end{bmatrix} = \begin{bmatrix} 180^\circ \\ -(25^\circ + 20^\circ - 20^\circ) \\ 25^\circ \end{bmatrix} = \begin{bmatrix} 180^\circ \\ -25^\circ \\ 25^\circ \end{bmatrix}$$

$$\mathbf{q}^{(3)} = \begin{bmatrix} -35^\circ & 35^\circ & -30^\circ & 30^\circ \end{bmatrix}^T$$

$$\rho_3 = 105.95 \cos(35^\circ) + 100 \cos(35^\circ - 30^\circ) = 186.408629 \text{ mm}$$

$$x_3 = 186.408629 \cos(-35^\circ) = 152.697009 \text{ mm} = 0.152697 \text{ m}$$

$$y_3 = 186.408629 \sin(-35^\circ) = -106.919597 \text{ mm} = -0.106920 \text{ m}$$

$$z_3 = 89.45 + 105.95 \sin(35^\circ) + 100 \sin(35^\circ - 30^\circ) = 158.935998 \text{ mm} = 0.158936 \text{ m}$$

$$\begin{bmatrix} \text{roll}_3 \\ \text{pitch}_3 \\ \text{yaw}_3 \end{bmatrix} = \begin{bmatrix} 180^\circ \\ -(35^\circ - 30^\circ + 30^\circ) \\ -35^\circ \end{bmatrix} = \begin{bmatrix} 180^\circ \\ -35^\circ \\ -35^\circ \end{bmatrix}$$

$$\mathbf{q}^{(4)} = \begin{bmatrix} 85^\circ & -20^\circ & 55^\circ & 25^\circ \end{bmatrix}^T$$

$$\rho_4 = 105.95 \cos(-20^\circ) + 100 \cos(-20^\circ + 55^\circ) = 181.475638 \text{ mm}$$

$$x_4 = 181.475638 \cos(85^\circ) = 15.816644 \text{ mm} = 0.015817 \text{ m}$$

$$y_4 = 181.475638 \sin(85^\circ) = 180.785068 \text{ mm} = 0.180785 \text{ m}$$

$$z_4 = 89.45 + 105.95 \sin(-20^\circ) + 100 \sin(-20^\circ + 55^\circ) = 110.570609 \text{ mm} = 0.110571 \text{ m}$$

$$\begin{bmatrix} \text{roll}_4 \\ \text{pitch}_4 \\ \text{yaw}_4 \end{bmatrix} = \begin{bmatrix} 180^\circ \\ -(-20^\circ + 55^\circ + 25^\circ) \\ 85^\circ \end{bmatrix} = \begin{bmatrix} 180^\circ \\ -60^\circ \\ 85^\circ \end{bmatrix}$$

$$\mathbf{q}^{(5)} = \begin{bmatrix} 80^\circ & -35^\circ & 55^\circ & -45^\circ \end{bmatrix}^T$$

$$\rho_5 = 105.95 \cos(-35^\circ) + 100 \cos(-35^\circ + 55^\circ) = 180.758421 \text{ mm}$$

$$x_5 = 180.758421 \cos(80^\circ) = 31.388370 \text{ mm} = 0.031388 \text{ m}$$

$$y_5 = 180.758421 \sin(80^\circ) = 178.012295 \text{ mm} = 0.178012 \text{ m}$$

$$z_5 = 89.45 + 105.95 \sin(-35^\circ) + 100 \sin(-35^\circ + 55^\circ) = 62.881591 \text{ mm} = 0.062882 \text{ m}$$

$$\begin{bmatrix} \text{roll}_5 \\ \text{pitch}_5 \\ \text{yaw}_5 \end{bmatrix} = \begin{bmatrix} 180^\circ \\ -(-35^\circ + 55^\circ - 45^\circ) \\ 80^\circ \end{bmatrix} = \begin{bmatrix} 180^\circ \\ 25^\circ \\ 80^\circ \end{bmatrix}$$